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Chapter 1
Basic Concepts

1.1 Introduction

Electric circuit theory and electromagnetic theory are the two fundamental theories upon which all branches of electrical engineering are built. Many branches of electrical engineering, such as power, electric machines, control, electronics, communications, and instrumentation, are based on electric circuit theory.

In electrical engineering, we are often interested in communicating or transferring energy from one point to another. To do this requires an interconnection of electrical devices, which we refer to as an electric circuit.

1.2 Systems of Units

As electrical engineers, we deal with measurable quantities. Our measurement, however, must be communicated in a standard language that virtually all professionals can understand, irrespective of the country where the measurement is conducted.

One great advantage of the SI unit is that it uses prefixes based on the power of 10 to relate larger and smaller units to the basic unit. Table 1.1 shows the SI prefixes and their symbols.

TABLE 1.1 The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10^{-3}	milli	m
10^{-6}	micro	μ

By using the SI units consistently, we can express any quantity in a compact form. For example, the following are expressions of the same distance in meters.

1.5 Power and Energy

Passive sign convention is satisfied when the current enters through the positive terminal of an element.

Although current and voltage are the two basic variables in an electric circuit, they are not sufficient by themselves. For practical purposes, we need to know how much power an electric device can handle. We all know from experience that a 100-watt bulb gives more light than a 60-watt bulb.

Power is the time rate of expending or absorbing energy, measured in watts (W).

We write this relationship as

$$p \triangleq \frac{dw}{dt} \tag{1.5}$$

where p is power in watts (W), w is energy in joules (J), and t is time in seconds (s). From Eqs. (1.1), (1.3), and (1.5), it follows that

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi \tag{1.6}$$

or

$$p = vi \tag{1.7}$$

The power p in Eq. (1.7) is a time-varying quantity and is called the instantaneous power. Thus, the power absorbed or supplied by an element is the product of the voltage across the element and the current through it. If the power has a + sign, power is being delivered to or absorbed by the element.

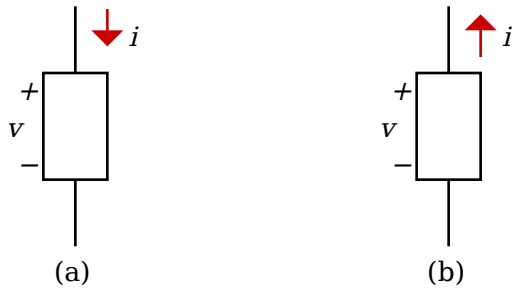


Figure 1.8 Reference polarities for power using the passive sign convention: (a) absorbing power, (b) supplying power.

When the voltage and current directions conform to Fig. 1.8(b), we have the active sign convention and p = −vi. Unless otherwise stated, we will follow the passive sign convention throughout this text.

In fact, the law of conservation of energy must be obeyed in any electric circuit. For this reason, the algebraic sum of power in a circuit, at any instant of time, must be zero:

$$\sum p = 0 \tag{1.8}$$

This again confirms the fact that the total power supplied to the circuit must balance the total power absorbed. From Eq. (1.6), the energy absorbed or supplied by an element from time t_0 to time t is

$$w = \int_{t_0}^{t} p\, dt = \int_{t_0}^{t} vi\, dt \tag{1.9}$$

Energy is the capacity to do work, measured in joules (J).

The electric power utility companies measure energy in watt-hours (Wh), where 1 Wh = 3,600 J.

Example 1.4

An energy source forces a constant current of 2 A for 10 s to flow through a lightbulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

Solution:

The total charge is

$$\not{q} = i\, \not{t} = 2 \times 10 = 20\ C$$

The voltage drop is

$$v = \frac{\not{w}}{\not{q}} = \frac{2.3 \times 10^3}{20} = 115\ V$$

Practice Problem 1.4

To move charge q from point a to point b requires 30 J. Find the voltage drop v_{ab} if: (a) q = 2 C, (b) q = 6 C.

Answer: (a) 15 V, (b) 5 V.